INCH-POUND

MIL-C-70949 (AR)
12 January 1990

# MILITARY SPECIFICATION

# CASE, CARTRIDGE, 105MM, M115B1E1

This Specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification contains requirements, quality assurance provisions, and packaging for the 105mm, M115B1E1 Spiral-Wrapped Cartridge Case (SWCC).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-46487 - Cleaning, Preparation and Organic Coating of Steel Cartridge Cases
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1315

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STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

7548235 - Cartons, Ammunition, for Interplant Shipment of Steel Cartridge Cases, Empty 37mm to 120mm, Inclusive
8663760 - Gage, Chamber
9323901 - Case, Cartridge, 105mm, M115B1E1

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ASTM Methods - AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The cartridge case shall comply with the requirements of the applicable drawings and referenced specifications.

3.2 Material. The supplier shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material used in manufacture.

3.3 Mechanical properties. The mechanical properties specified on drawings and referenced specifications shall be uniform throughout each section.

3.4 Protective coating. Protective coating as specified on drawings and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration and shall be uniform and continuous. Scratches exposing base metal are not permitted.

3.5 Assembly tightness. The case shall be assembled in such a manner that there is no relative rotation between the base, the cup, and the body when a torque of 75±5 in.-lbs. is applied to the base.

3.6 Maximum profile and alignment. Maximum profile and alignment of the cartridge case shall comply with chamber gage drawing 8663760.

3.7 Ballistics. The cartridge case shall comply with the requirements of 3.7.1 through 3.7.5 when subjected to ballistic testing in accordance with 4.6.7 in the following phases:

a. Excess pressure phase
b. Cold temperature phase
c. Excess, excess pressure phase (for information only).

3.7.1 Obturation. The cartridge case shall effectively seal the gun breech against leakage of gases generated during firing.



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3.7.2 Extraction. After firing, the cartridge case shall be extracted by the service extractors when the gun is operated in the normal manner.

3.7.3 Primer retention. The cartridge case shall retain its primer during firing and extraction.

3.7.4 Propellant gas leakage. The cartridge case shall effectively prevent the entry of propellant gas through or between case components into the gun chamber.

3.7.5 Cracks or splits. After firing, the cartridge case shall be free of cracks or splits through all body plies.

3.8 Workmanship. The requirements for workmanship are as specified on drawings, referenced specifications and the following:

3.8.1 Metal defects. The cartridge case shall be free of cracks, blisters, laps, deep scratches, dents, embedded foreign matter, porosity, and other metal defects.

3.8.2 Burrs. No cartridge case shall have any burrs.

3.8.3 Foreign matter. No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.



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4.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to accept defective material.

4.3 Classification of inspections. The following types of inspections shall be performed on this item:

a. First article inspection

b. Quality Conformance Inspection

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of this specification. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency, shall also be submitted whenever a change occurs in specification, or source of supply, such as to significantly affect product uniformity as determined by the responsible government technical agency. The first article sample shall consist of the following items in quantities indicated.

| Part Description | Drawings | Quantity |
|---|---|---|
| Cup (uncoated) | 11735732 | 4 |
| Cup (coated) | 11735732 | 10 |
| Collar (uncoated) | 11735733 | 4 |
| Collar (coated) | 11735733 | 10 |
| Base (uncoated) | 11735735 | 4 |
| Base (coated) | 11735735 | 10 |
| Test Panels (See MIL-C-46487) | -------- | 2 |
| Cartridge Case Body Blank | -------- | 4 |
| Cartridge Case (M115B1E1) | 9323901 | 254 |

4.4.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.4.3 Rejection. See MIL-A-48078



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Table I - First article inspection

## CLASSIFICATION OF CHARACTERISTICS

MIL-C-70949 (AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| | M115B1E1 Cartridge Case Assembly | 1 OF 2 | See Below |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| | Cup (uncoated)<br>Dwg. 11735732<br>Hardness | 4 units | 3.3 | |
| | Collar (uncoated)<br>Dwg. 11735733<br>Hardness | 4 units | 3.3 | |
| | Base (uncoated)<br>Dwg. 11735735<br>Hardness | 4 units | 3.3 | |
| | Cup (coated)<br>Dwg. 11735732<br>Examination for defects<br>Coating thickness<br>Salt spray | 10 units<br>4 units<br>2 units | 3.1<br>3.4<br>3.4 | |
| | Collar (coated)<br>Dwg. 11735733<br>Examination for defects<br>Coating thickness<br>Salt spray | 10 units<br>4 units<br>2 units | 3.1<br>3.4<br>3.4 | |

NOTES:



AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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Table I - First article inspection

## CLASSIFICATION OF CHARACTERISTICS

MIL-C-70949 (AR)

| PARAGRAPH | TITLE | SHEET 2 OF 2 | DRAWING NUMBER |
|---|---|---|---|
| | M115B1E1 Cartridge Case Assembly | | See Below |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| | Base (coated) | | | |
| | Dwg. 11735735 | | | |
| | Examination for defects | 10 units | 3.1 | |
| | Coating thickness | 4 units | 3.4 | |
| | Salt spray | 2 units | 3.4 | |
| | Test panels (see MIL-C-46487) | | | |
| | Weight of phosphate coating | 2 units | 3.4 | |
| | Flexibility | 2 units | 3.4 | |
| | Cartridge case body blank | | | |
| | Hardness | 4 units | 3.3 | |
| | Coating thickness | 4 units | 3.4 | |
| | Salt spray | 2 units | 3.4 | |
| | Cartridge case | | | |
| | Dwg. 9323901 | | | |
| | Examination for defects | 254 units | 3.1 | |
| | Distance from base O.D. to edge of body plies | 2 units | 3.1 | |
| | Salt spray | 2 units | 3.4 | |
| | Ballistics test | | 3.7 | |
| | Excess pressure | 100 units | | |
| | Cold temperature | 50 units | | |
| | Excess, excess pressure (info only) | 100 units | | |

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4.5 Quality conformance inspection.

4.5.1 Lot formation. Lots shall comply with the lot formation provisions of MIL-A-48078. In addition one lot of cartridge cases shall consist of not more than 25,000 units.

4.5.2 Examinations and tests.

4.5.2.1 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Where specified herein, attributes sampling inspection shall be conducted in accordance with Table II below. Samples shall be selected at random.

Table II. Sampling

| Lot Size | Inspection Level | |
|---|---|---|
| | I Major Defects | II Minor Defects |
| 2 to 13 | * | * |
| 14 to 90 | * | 13 |
| 91 to 150 | 125 | 13 |
| 151 to 500 | 125 | 30 |
| 501 to 1200 | 125 | 35 |
| 1201 to 3200 | 125 | 43 |
| 3201 to 10000 | 125 | 50 |
| 10001 to 35000 | 296 | 61 |

Numbers under inspection levels indicate sample size.

* Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for both inspection levels.



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4.5.2.2 Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.5.3 Tool control dimensions. Dimensions marked "Tool Control" shall be gaged at the beginning or end of each (24 hour) working day, on each machine in operation and for the first component produced after any tooling change, adjustment or repair. When destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.5.4 Inspection equipment. Inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the tables starting with paragraph 4.5.5.

4.5.5 Classification of characteristics.



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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.1 | Cup | 1 OF 1 | 11735732 |
| | | | NEXT HIGHER ASSEMBLY 9323901 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Outside diameter | Table II | 3.1 | Gage |
| 102 | Inside diameter | Table II | 3.1 | Gage |
| 103 | Length, total | Table II | 3.1 | Gage |
| 104 | Diameter of recess | Table II | 3.1 | Gage |
| 105 | Angle of recess | Table II | 3.1 | Gage |
| 106 | Thickness of lip | Table II | 3.1 | Gage |
| 107 | Taper on diameter | Table II | 3.1 | Gage |
| 108 | Hardness | 4.5.6.1 | 3.3 | 4.6.1.3 |
| 109 | Thickness | Table II | 3.1 | Gage |
| 110 | Protective coating missing or inadequate | Table II | 3.4 | Visual |
| 111 | Surface finish | Table II | 3.1 | Visual |
| 112 | True position of I.D. | Table II | 3.1 | Gage |
| 113 | Radius at corner | Table II | 3.1 | Gage |
| 114 | Coating thickness | 4.5.6.2.1 | 3.4 | Gage |
| 115 | Salt spray | 4.5.6.2.2 | 3.4 | 4.6.2.1 |
| Minor | | | | |
| 201 | Poor workmanship | Table II | 3.8 | Visual |

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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.2 | Collar | 1 OF 1 | 11735733 |
| | | | NEXT HIGHER ASSEMBLY: 9323901 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Boss outside diameter | Table II | 3.1 | Gage |
| 102 | Width of boss | Table II | 3.1 | Gage |
| 103 | Wall thickness of boss (two places) | Table II | 3.1 | Gage |
| 104 | Depth of boss groove | Table II | 3.1 | Gage |
| 105 | Large outside diameter | Table II | 3.1 | Gage |
| 106 | Circumferential taper angle of boss | Table II | 3.1 | Gage |
| 107 | Circumferential taper depth of boss | Table II | 3.1 | Gage |
| 109 | Radius | Table II | 3.1 | Gage |
| 110 | True position of outside diameter | Table II | 3.1 | Gage |
| 111 | Large radius of flange | Table II | 3.1 | Gage |
| 112 | Radius between boss and flange | Table II | 3.1 | Gage |
| 113 | Hardness | 4.5.6.1 | 3.3 | 4.6.1.2 |
| 114 | Protective coating missing or inadequate | Table II | 3.4 | Visual |
| 115 | Surface finish | Table II | 3.1 | Visual |
| 116 | Coating thickness | 4.5.6.2.1 | 3.4 | Gage |
| 117 | Salt spray | 4.5.6.2.2 | 3.4 | 4.6.2.1 |
| Minor | | | | |
| 201 | Small radius of flange | Table II | 3.1 | Gage |
| 202 | Poor workmanship | Table II | 3.8 | Visual |

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## CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.3 | Base | 1 OF 2 | 11735735 |
| | | | NEXT HIGHER ASSEMBLY 9323901 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Large outside diameter | Table II | 3.1 | Gage |
| 102 | Small outside diameter | Table II | 3.1 | Gage |
| 103 | True position of small outside diameter | Table II | 3.1 | Gage |
| 104 | Diameter of outside edge of channel | Table II | 3.1 | Gage |
| 105 | True position of outside edge of channel diameter | Table II | 3.1 | Gage |
| 106 | Width of channel | Table II | 3.1 | Gage |
| 107 | Angle of channel walls | Table II | 3.1 | Gage |
| 108 | Pitch diameter of thread, max. | Table II | 3.1 | Gage |
| 109 | Minor diameter of thread, max. | Table II | 3.1 | Gage |
| 110 | True position of pitch diameter | Table II | 3.1 | Gage |
| 111 | Diameter of thread relief | Table II | 3.1 | Gage |
| 112 | True position of thread relief diameter | Table II | 3.1 | Gage |
| 113 | Diameter of counterbore | Table II | 3.1 | Gage |
| 114 | True position of counterbore diameter | Table II | 3.1 | Gage |
| 115 | Depth of thread relief | Table II | 3.1 | Gage |
| 116 | Depth of counterbore | Table II | 3.1 | Gage |
| 117 | Distance from flange to bottom of counterbore | Table II | 3.1 | Gage |
| 118 | Height of channel protrusion | Table II | 3.1 | Gage |
| 119 | Width of channel protrusion | Table II | 3.1 | Gage |

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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.3 | Base | 2 OF 2 | 11735735 |
| | | | NEXT HIGHER ASSEMBLY 9323901 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Major | (cont'd) | | | |
| 120 | Overall thickness of base | Table II | 3.1 | Gage |
| 121 | Depth of channel | Table II | 3.1 | Gage |
| 122 | Thickness of flange | Table II | 3.1 | Gage |
| 123 | Hardness | 4.5.6.1 | 3.3 | 4.6.1.1 |
| 124 | Surface finish | Table II | 3.1 | Visual |
| 125 | Protective coating missing or inadequate | Table II | 3.4 | Visual |
| 126 | Coating thickness | 4.5.6.2.1 | 3.4 | Gage |
| 127 | Salt spray | 4.5.6.2.2 | 3.4 | 4.6.2.1 |
| 128 | Metal defects | 100% | 3.8.1 | 4.6.3 |
| Minor | | | | |
| 201 | Max. offset of the diameter of the outside surface of the channel at 45° | Table II | 3.1 | Gage |
| 202 | Max. offset of the diameter of the inside surface of the groove at 45° | Table II | 3.1 | Gage |
| 203 | Radii or chamfers missing | Table II | 3.1 | Visual |
| 204 | Poor workmanship | Table II | 3.8 | Visual |

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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.4 | M115B1E1 Cartridge Case Assembly | 1 OF 3 | 9323901 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined | | | |
| Major | | | | |
| 101 | Thickness of body material prior to wrapping | Table II | 3.1 | Gage |
| 102 | Total length | Table II | 3.1 | Gage |
| 103 | Distance from flange (base) to first basic diameter | Table II | 3.1 | Gage |
| 104 | Distance from flange (base) to second basic diameter | Table II | 3.1 | Gage |
| 105 | Distance from flange (base) to third basic diameter | Table II | 3.1 | Gage |
| 106 | Distance from flange (base) to fourth basic diameter | Table II | 3.1 | Gage |
| 107 | Inside diameter of mouth | Table II | 3.1 | Gage |
| 108 | True position of mouth I.D. | Table II | 3.1 | Gage |
| 109 | O. D. of mouth | Table II | 3.1 | Gage |
| 110 | True position of outside diameter of mouth | Table II | 3.1 | Gage |
| 111 | Improper number of wraps at mouth | Table II | 3.1 | Visual |
| 112 | Protective coating missing or inadequate | Table II | 3.4 | Visual |
| 113 | Minimum radius of sheet edge | Table II | 3.1 | Visual |
| 114 | Location of oval indents from mouth | Table II | 3.1 | Gage |

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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF CHARACTERISTICS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.4 | M115B1E1 Cartridge Case Assembly | 2 OF 3 | 9323901 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Major | (cont'd) | | | |
| 115 | Angular location of oval indents to one another | Table II | 3.1 | Gage |
| 116 | Location of weld to edge of inner wrap | Table II | 3.1 | Gage |
| 117 | Length of each oval indent | Table II | 3.1 | Gage |
| 118 | Width of each oval indent | Table II | 3.1 | Gage |
| 119 | Hardness of body blank | 4.5.6.1 | 3.3 | 4.6.1.4 |
| 120 | Assembly tightness | 100% | 3.5 | 4.6.5 |
| 121 | Metal defects | 100% | 3.8.1 | 4.6.3 |
| 122 | Salt spray (body blank) | 4.5.6.2.2 | 3.4 | 4.6.2.1 |
| 123 | Salt spray (assembly) | 4.5.6.2.2 | 3.4 | 4.6.2.1 |
| 124 | Distance from base O.D. to outer ply | 4.5.6.3 | 3.1 | 4.6.6 |
| 125 | Distance from base O.D. to two inner plies | 4.5.6.3 | 3.1 | 4.6.6 |
| 126 | Radii of single wrap circular indents | Table II | 3.1 | Gage |
| 127 | Locations of single wrap circular indents from mouth edge | Table II | 3.1 | Gage |
| 128 | Depth of single wrap circular indents | Table II | 3.1 | Gage |
| 129 | Locations of double wrap circular indents from mouth edge | Table II | 3.1 | Gage |
| 130 | Radii of double wrap circular indents | Table II | 3.1 | Gage |
| 131 | Depth of double wrap circular indents | Table II | 3.1 | Gage |

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## CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.5.5.4 | M115B1E1 Cartridge Case Assembly | 3 OF 3 | 9323901 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Major | (cont'd) | | | |
| 132 | Location of weld to edge of outer wrap | Table II | 3.1 | Gage |
| 133 | Profile and alignment | 100% | 3.6 | Gage |
| 134 | Ballistics | 4.5.6.4.2 | 3.7 | 4.6.7.2 |
| Minor | | | | |
| 201 | Radii missing | Table II | 3.1 | Visual |
| 202 | Stamping incorrect or illegible | Table II | 3.1 | Visual |
| 203 | Location of spot welds from mouth | Table II | 3.1 | Visual |
| 204 | 45° overlap of body wraps at mouth | Table II | 3.1 | Gage |
| 205 | Poor workmanship | Table II | 3.8 | Visual |

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4.5.6 Testing. This section describes sampling plans, criteria for acceptance or rejection of individual samples, and criteria for acceptance or rejection of the entire lot.

4.5.6.1 Hardness. Ten samples shall be randomly selected for testing. Failure of one hardness reading is cause for rejection of the sample. Failure of one sample is cause for rejection of the lot.

4.5.6.2 Protective coating.

4.5.6.2.1 Coating thickness. Samples shall be selected in accordance with Table II. Failure to comply with the thickness requirement in the component drawing shall be cause for rejection of the sample. Failure of one sample shall be cause for rejection of the lot.

4.5.6.2.2 Salt spray. Five samples shall be randomly selected for testing. Samples shall be evaluated at the end of the test period in accordance with the detail specification for the protective finish. The cartridge case assemblies subjected to the salt spray test shall show no more than 1/8 inch creepage, blistering, or loss of adhesion of the coating from the scribe mark. At all other points, there shall be no more than a trace of film failure (rust grade 9 ASTM D-610), and not more than 5 scattered blisters none larger than 1mm (3/64 inch) in diameter on a 48 square inch area. Rust within one thirty second (1/32) inch of all other edges shall be permitted. Failure to comply shall be cause for rejection of the sample. Rejection of one sample shall be cause for rejection of the lot.

4.5.6.3 Distance from base O.D. to edge of body plies. In order to determine the integrity of the assembly, two samples shall be inspected during each shift or eight hour production period. One sample shall be from the middle and one from the end of the shift. If any change is made to the assembly process, another sample shall be inspected. Failure of any sample to comply with requirements shall be cause for rejection of all production subsequent to the last acceptable inspection.

4.5.6.4 Ballistics. Physical evidence of non-compliance with the applicable table below, with or without video indications is cause for rejection. Video detection of emissions from the breach area alone is insufficient cause for rejection.



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4.5.6.4.1 First article. The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III.

Table III

Acceptance Criteria For First Article Ballistics Test

| Defect Characteristic | Accept | Reject |
|---|---|---|
| 1. Obturation failure | 0 | 1 |
| 2. Extraction failure | 0 | 1 |
| 3. Primer retention failure | 0 | 1 |
| 4. Propellant gas leakage (Base burn mark) | 0 | 1 |
| 5. Propellant gas leakage (Cup burn mark) | 0 | 1 |
| 6. Propellant gas leakage (Body burn mark) | 3 | 4 |
| 7. Cracks or splits through all Body plies | 3 | 4 |



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4.5.6.4.2 Production lot. One hundred cartridge cases from each lot will be ballistically tested in the excess pressure phase. Acceptance and rejection criteria shall be in accordance with Table IV. If test firing experiences an extraction or primer retention failure, the lot is eligible for retest firing of an additional sample of two hundred cases. The cumulative sample of three hundred must now meet the requirements of Table IV.

Table IV

Acceptance Criteria For Production Lot Ballistics Test

| Defect Characteristic | First Submission Accept | First Submission Reject | Cumulative Submission Accept | Cumulative Submission Reject |
|---|---|---|---|---|
| 1. Obturation failure | 0 | 1 | Not permitted | |
| 2. Extraction failure | 0 | 2 | 1 | 2 |
| 3. Primer retention fail. | 0 | 2 | 1 | 2 |
| 4. Propellant gas leakage (Base burn mark) | 0 | 1 | Not permitted | |
| 5. Propellant gas leakage (Cup burn mark) | 0 | 1 | Not permitted | |
| 6. Propellant gas leakage (Body burn mark) | 2 | 3 | Not permitted | |
| 7. Cracks or splits through all Body plies | 2 | 3 | Not permitted | |

4.6 Methods of inspection.

4.6.1 Hardness test. Hardness tests shall be conducted on components in accordance with ASTM E18 prior to cartridge case assembly.

4.6.1.1 Base. Seven hardness values shall be taken across the center of the rear surface of the base on both sides of the primer hole at approximately 1/4 inch intervals.

4.6.1.2 Collar. Four hardness values shall be taken on the underside of the flange of the collar approximately 90° apart.

4.6.1.3 Cup. Four hardness values shall be taken on the cylindrical surface of the cup, excluding the flange, approximately 90° apart.



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4.6.1.4 Body blank. Hardness values shall be taken at approximately 12 inch intervals both vertically and horizontally through the center of the length and width of the body blank.

4.6.2 Salt spray. Salt spray test shall be performed utilizing procedures specified in ASTM B117 and in accordance with the applicable detailed specification for the protective finish, except the salt exposure periods and solution shall be in accordance with the component drawings.

4.6.3 Metal defects detection. Four power (4X) magnification shall be used in the examination of the base and the exterior surface of the case assembly. The interior of the case assembly may be examined without magnification.

4.6.4 Maximum profile and alignment. Testing shall be conducted after application of protective coating. The case shall freely enter a maximum profile and alignment gage, Dwg. 8663760. The gate of the gage shall close without interference with the cartridge case base.

4.6.5 Assembly tightness. The cartridge case shall be held in an appropriate fixture and 75±5 in-lbs of torque shall be applied in both directions between the base and body. The moment arm shall be normal to the centerline of the case. Observation shall be made for any relative motion between the base, the cup and the body.

4.6.6 Distance from base O.D. to edge of body plies. Samples shall be cut along a center line of the case and gaged for minimum positioning depth of all body plies between the collar and cup. The sectioned case shall also be inspected for evidence of metal defects and poor workmanship.

4.6.7 Ballistics test. A color video camera shall be aimed at the breach of the gun and operated in such a manner as to detect any evidence of propellant gas leakage during firing. Video shall be used as an aid in detection of propellant gas leakage and inadequate obturation by identifying suspect cases. Video recordings will not be used as accept reject criteria. Camera exposure rate shall be not less than 30 frames per second. Provisions such as metal or canvas drapes shall be in place to shield the camera from blinding caused by muzzle flash. To prevent the recording and interpretation of extraneous video signals as propellant gas leakage during ballistic testing, the gun must be free of all lubricants and rust preventives in the breech and chamber areas. If such materials have been previously applied, the gun shall be wiped clean and warmer rounds fired to clear any residue. All cases shall be examined immediately after firing for compliance with defect characteristics listed in Tables III or IV as applicable.



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4.6.7.1 First article sample.

4.6.7.1.1 Test phases.

4.6.7.1.1.1 Excess pressure phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure with the following limits. Minimum mean value of the test rounds fired to be 64,000 psi (441.26 MPa) piezo equivalent pressure and a maximum individual pressure to be 68,500 psi (472.29 MPa) piezo equivalent pressure. The complete rounds will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5°C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.1.2 Cold temperature phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 57,100 psi (393.69 MPa) to 60,600 psi (417.82 MPa) piezo equivalent pressure, when temperature conditioned at 70°F. The complete rounds will be temperature conditioned at -35°F + 5°F (-37°C + 3°C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.1.3 Excess, excess pressure phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 72,300 psi (498.49 Mpa) to 74,200 psi (511.59 Mpa) piezo equivalent pressure. The cartridge will be will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5 °C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.4 Failure modes.

4.6.7.1.4.1 Obturation. The case shall be examined for evidence of obturation failure, i.e. failure of the cup to expand sufficiently to prevent passage of propellant gas to the rear of the chamber enclosing the case base. This failure is evidenced by presence of burn mark(s) on the cup or base.

4.6.7.1.4.2 Extraction. Cartridge cases which failed to completely eject shall be examined for sufficient looseness in the gun chamber to permit removal by hand.

4.6.7.1.4.3 Primer retention. The case shall be examined for retention of its primer after firing.



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4.6.7.1.4.4 Propellant gas leakage. The cartridge case shall be examined for entry of propellant gas into the gun chamber by passing through any component or component interface. Mechanisms of failure include but are not limited to burn-through, rupture, joint failure, cracks and gas flow between body plies. Leakage is evidenced by the presence of burn marks on the component causing failure.

4.6.7.1.4.5 Cracks or splits. The cartridge case shall be visually examined for cracks or splits.

4.6.7.2 Production lot. Production lot acceptance tests shall be conducted in the Excess Pressure Phase. See 4.6.7.1.1.1 herein.

5. PACKAGING

5.1 Preservation and packaging. Packaging shall be as specified on Dwg. 7548235.

5.2 Packing and marking. Packing and marking shall be as specified on Dwg. 7548235 and MIL-STD-1169.

6. NOTES. (This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This cartridge case is intended to be used in the 105mm, target practice tank ammunition.

6.2 Process control lots. Prior to fabrication of a first article sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to five process control lots. These process control lots shall consist of not less than ten nor more than twenty-five cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of these lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.3 Visual examination qualification. When compliance with requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.4 Ordering data. See MIL-A-48078.



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6.5 Submission of firing records and ammunition data cards. In addition to the normal distribution of records, when the cartridge case is procured by the Army Armament, Munitions and Chemical Command, one copy of all firing records and ammunition data cards shall be forwarded to:

Commander
U. S. Army Armament Research, Development, and Engineering Center
Attn: AMSMC-QAT-A (D)/SMCAR-CCH-P
Picatinny Arsenal, NJ 07806-5000

6.6 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to:

Commander
U. S. Army Armament Research, Development, and Engineering Center
Attn: AMSMC-QAT-I (D)
Picatinny Arsenal, NJ 07806-5000

6.7 Subject term (key word) listing.

Ammunition
Tank
Cannon
Cartridge
Case, Cartridge
Spiral-Wrapped

Custodian:
ARMY-AR

Preparing activity:
ARMY-AR

Project No: 1315-AC07



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-70949 (AR) | CASE, CARTRIDGE, 105MM, M115B1E1 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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